Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping

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Hiring Team *12 November 2026*

NCS Multistage
Calgary, AB

Project Engineer / Leader (Level I)

Dear NCS Multistage Hiring Team,

I recently completed my MASc in Mechanical and Mechatronics Engineering at the University of Waterloo and am currently based in Calgary. I'm interested in NCS Multistage's Project Engineer / Leader position because of the opportunity to work on mechanical design, prototyping and testing. Through my projects and work experience, I've enjoyed the practical side of engineering, especially designing and building test equipment and evaluating how well it performs.

I've gained much of my practical design experience through building test equipment. During my UBC capstone, I developed silicone and agar bladder phantoms and built the control and data acquisition interface for a pressurised test system. The system monitored pressure and volume in real time and performed automated fill and void cycles. I validated stable, repeatable pressurisation at 2.2 kPa and tested up to 4.75 kPa. I continued working on the project with Echoscope, where I designed and iterated the test platform PCB and wrote five manufacturing and verification procedures covering assembly, wiring, software setup and functional testing. All five procedures were approved and used by the client.

During my internship at GN Corporation in Airdrie, I prepared work instructions and SOPs based on engineering drawings and customer specifications. I also reduced fibreglass cutting waste by 50% through process analysis. During my graduate research, I validated an LS-DYNA model against physical sled tests and investigated why the simulations and tests produced different results. I've documented my research through technical reports, my thesis, a conference paper and six sponsor presentations. I'm also comfortable using SolidWorks, 3D printing, a lathe, mill and drill press.

I'm currently based in Calgary, eligible to register as an EIT with APEGA and available to start immediately. I'm comfortable with shop and field work and would like to learn more about well completions and downhole tools. I'd appreciate the opportunity to discuss the position further and how my experience could contribute to the team.

Sincerely,

Patrick Thangarajah